JAYASHREE SIVAKUMAR

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Career Objective

I am passionate about **VLSI** and **semiconductor technology**, and I want to contribute to real-world semiconductor projects by applying my knowledge of **Digital Electronics, Verilog HDL, RTL Design, FSM, and Testbench** development. I am eager to learn industry practices and continuously enhance my technical expertise.

Technical Skills

HDL: Verilog
VLSI: RTL Coding, FSM-Based Design, ASIC Design Flow
Protocols: AMBA AHB, APB, AHB-to-APB Bridge
EDA Tools: Xilinx Vivado, ModelSim, Quartus Prime, EDA Playground
Embedded/IoT: ESP32, Arduino UNO, ThingSpeak, RTC Module, Servo Motor, IR Sensor
Digital Design: Combinational & Sequential Logic, FSM, CMOS Fundamentals
Programming: C, Python

Internship

VLSI Design Internship – Maven Silicon Softech Pvt Ltd, Bengaluru *June – July 2026 / 45 days*

- Built a strong foundation in **Digital Electronics**, including logic gates, Boolean algebra, combinational circuits, and sequential circuits.
- Designed and implemented a **Finite State Machine (FSM)** for digital control systems.
- Developed RTL design skills using **Verilog HDL** -writing synthesis modules and testbenches with hardware-oriented coding practices.
- Gained practical knowledge of functional simulation with **ModelSim** and RTL synthesis with **Quartus Prime**.
- Strengthened understanding of **RTL design flow, protocol conversion, pipelining, address decoding** and digital hardware implementation

Projects

AHB to APB Bridge Protocol *July 2026*

- Designed and integrated RTL modules, including **AHB Slave Interface, APB Controller, APB interface, Bridge Top and Testbench**.
- Implemented **protocol conversion** between AHB and APB buses, enabling communication between high-speed and low-speed peripherals.
- Developed an **FSM-based APB controller** to manage read and write transactions as per the APB protocol specification.
- Implemented **address pipelining, write data pipelining, address decoding**, and valid transaction detection for reliable bus communication.
- Verified the RTL design through **functional simulation** using **ModelSim** and performed **hardware synthesis** using **Quartus Prime**.

IoT-Based Weather Monitoring System using ESP32 & ThingSpeak *May 2026*

- Built a weather monitoring system using **ESP32** to measure and log real-time temperature, humidity, and pressure data.
- Integrated with the **ThingSpeak IoT cloud platform** to visualise sensor data through live charts and enable remote monitoring from any device.
- Configured **ESP32 Wi-Fi connectivity** for continuous data transmission to the cloud at defined intervals.

- **Tools & Technologies:** ESP32, ThingSpeak, DHT Sensors, Embedded C.

IoT-Based Smart Medicine Reminder & Dispenser *December 2025*

- Designed a smart healthcare system using **ESP32, RTC module, servo motor, buzzer, and IR sensor** to remind users and automatically dispense the required medicine dose at scheduled times.
- Integrated the system with the **Blynk IoT platform** to send real-time notifications and store medicine intake data, enabling patients and caregivers to monitor medication remotely.
- Implemented **dose scheduling and dispensing logic** on Arduino Uno with servo-controlled compartment rotation triggered by RTC alarm.
- Used an **IR sensor** for dose detection and a **buzzer alert** for missed-dose notification, ensuring reliable and accurate medication management.
- **Tools & Technologies:** ESP32, Arduino Uno, Blynk, RTC Module, Servo Motor, IR Sensor, C/C++.

Education

B.E. – Electronics and Communication Engineering *2027*
Government College of Engineering, Erode CGPA: 7.7

Behavioural Skills

- **Analytical thinking** with structured problem decomposition — applied in hardware design and algorithm implementation.
- **Self-motivated learner** with the ability to quickly adapt to new tools, protocols, and design methodologies.
- **Collaborative team player** with clear technical communication skills, presentation and monitoring.
- **Persistent and detail-oriented** — reflected in simulation debugging and waveform analysis practices.

Languages

Tamil, English, Kannada

Achievements

- **Runner-up**, **AI for Sustainability Hackathon** - demonstrated cross-domain innovation and problem-solving under competitive conditions.
- Selected up to **Round 3**, **Naan Mudhalvan Data Science Hackathon** (Tamil Nadu Government Skill Development Initiative) – presented a project on a Colour Detection tool.